

Quantitative analysis in graduate research:

a quick guide

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A little about me

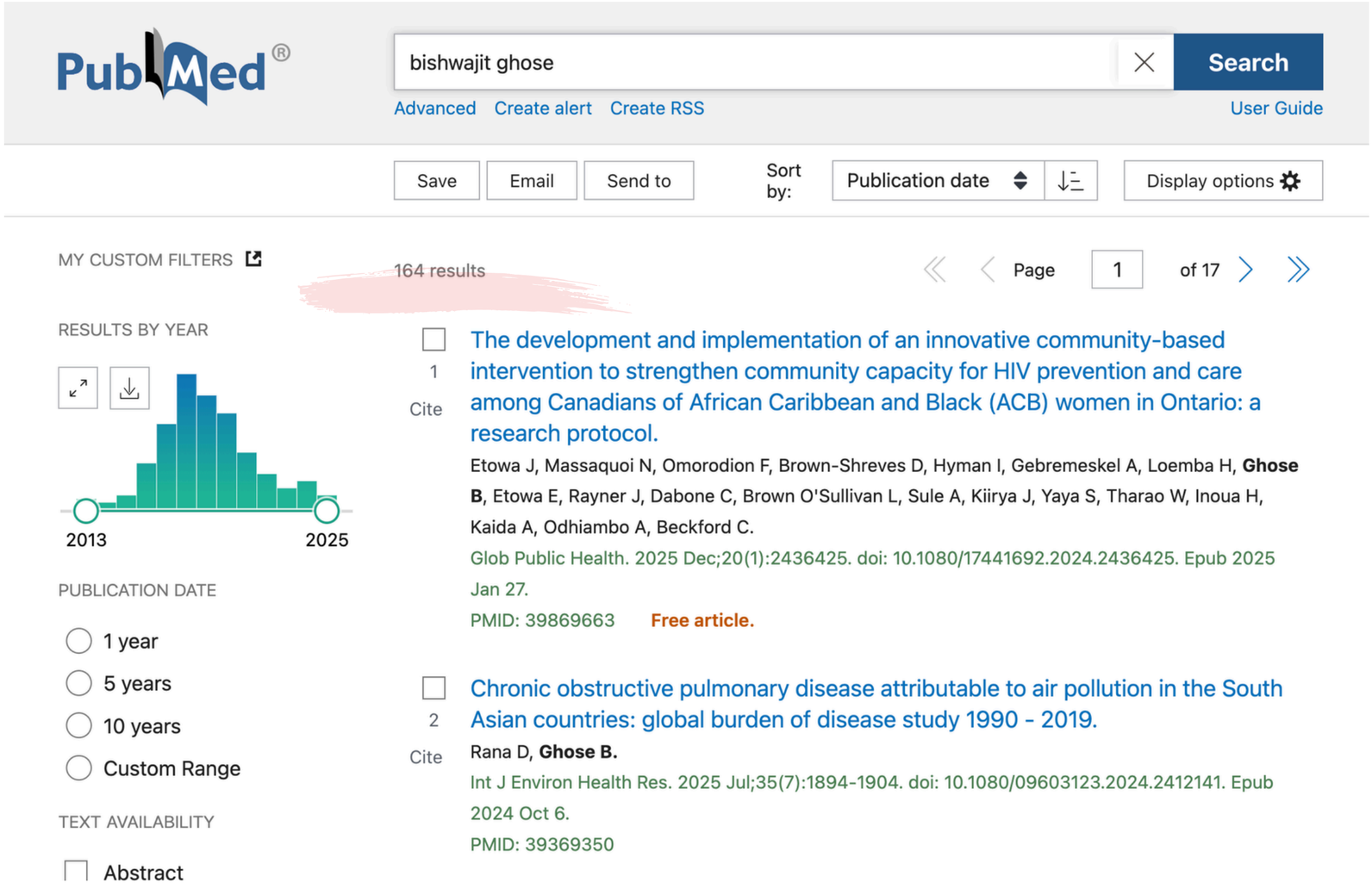
PhD, Epidemiology
MS, Big Data
MPH, Nutrition

Research interest

Quantitative analysis
Data viz, Natural Language Processing

Research tools

R, Stata
Python, Tableau



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CONTENT

- Basics of quantitative research
- Sources of data for quantitative research
- Key steps in data analysis
- Types of variables
- Types of study designs
- Types of analytical methods

BY THE END OF THE LECTURE YOU SHOULD BE ABLE TO…

- Understand the role of study designs in research
- Understand the role of context in research
- Develop clear and effective research topics/questions.
- Learn the types of variables
- Choose appropriate analytical methods
- Correctly interpret results
- Start analysing data for your thesis/project

In God we trust.

All others must bring data.

W. Edwards Deming

Basics of Quantitative research

Quantitative research: key characteristics

- Uses ~~numeric~~ quantifiable data
- Employs statistical methods to explore patterns/ tests hypotheses/establish cause-and-effect relationships
- Powerful for measuring, benchmarking, and generalizability.
- main strengths are objectivity (!) and precision.
- main weakness are lack of depth and context.
- main challenges are poor design and misinterpretation.
- not a standalone solution (works best in combination with qualitative research)

Quantitative research

- A systematic way of asking questions about the world, designing a study, gathering measurable evidence, and testing hypotheses.
- It covers the full process: formulating hypotheses, designing surveys or experiments, processing data, and interpreting results in context.

Statistics

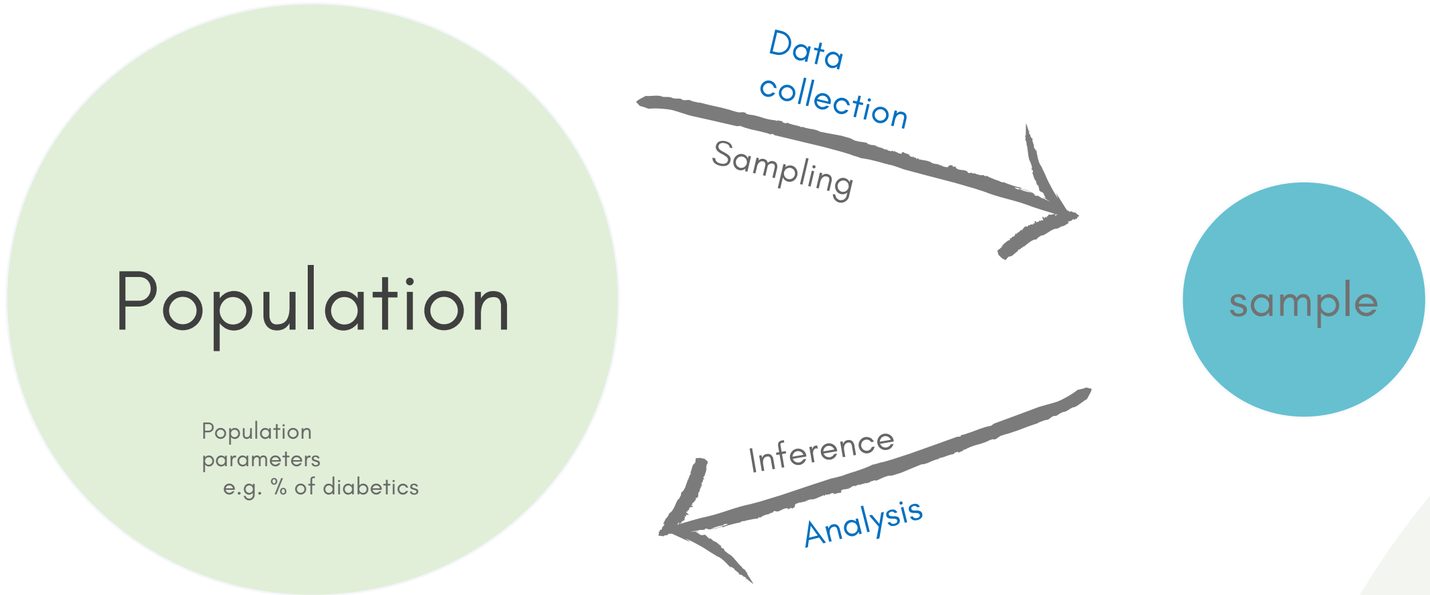
- A set of mathematical tools and theories for summarizing, analyzing, and drawing conclusions from data.
- Deals with probability, inference, significance testing, modeling, uncertainty.

Data analysis

- The practical process of working with data e.g. collecting, cleaning and analysing to explore patterns.
- It sits between raw data and insight, using both statistical techniques and domain knowledge.

Statistical inference

involves 2 steps: 1) Data collection & 2) estimation



Sources of data for quantitative research

Primary data

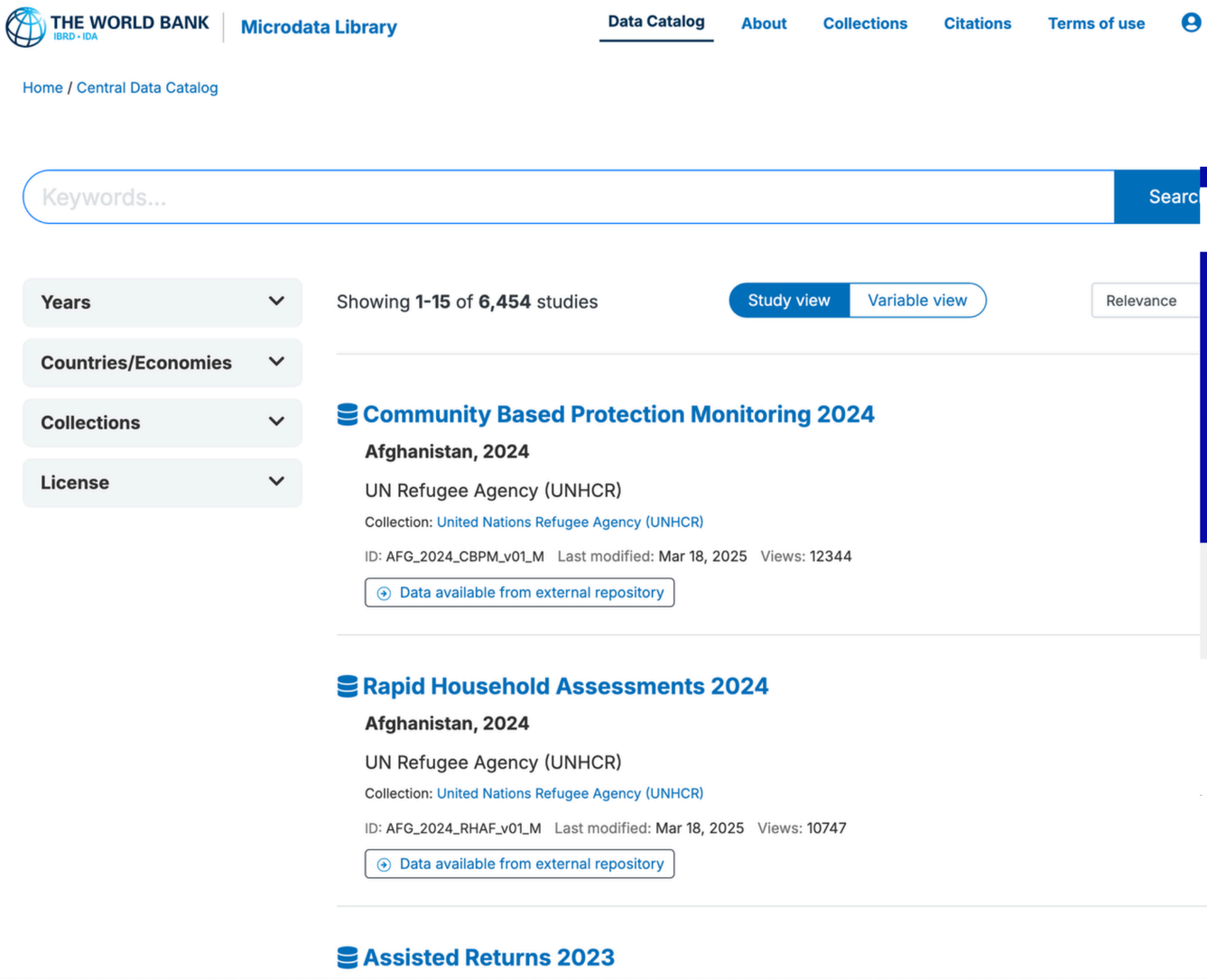
Data collected directly by the researcher to address their specific research problem. It is original and unprocessed.

- Researcher controls the design, methodology, and variables.
- Tailored specifically to the research question.
- Typically more expensive and time-consuming to gather.

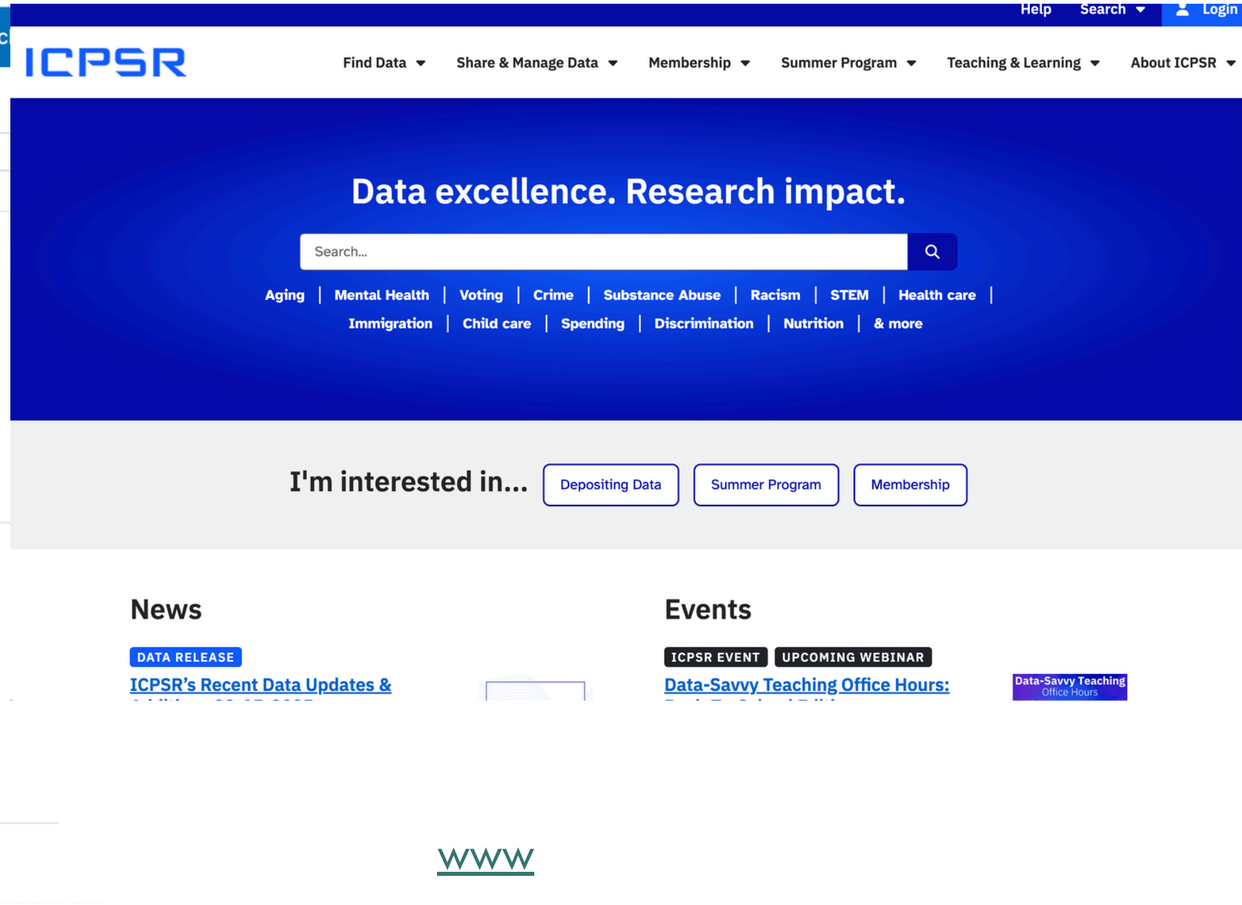
Secondary data

Data that was collected by someone else for a different purpose but is repurposed for a new research question.

- Researcher has no control on the design, methodology, and variables.
- Often provides access to large-scale data.
- May not perfectly align with the current research needs; requires evaluation of source credibility.

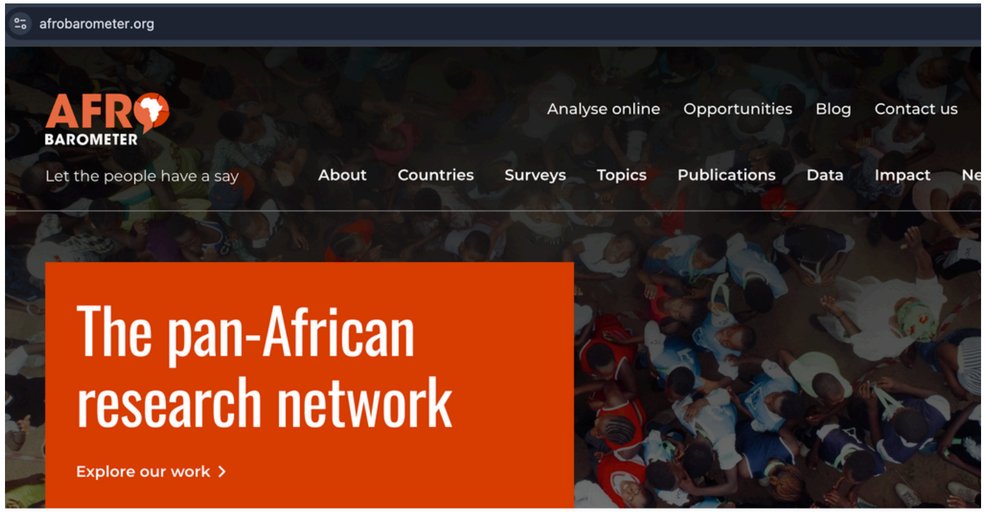


WWW



WWW

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AFRO BAROMETER

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Home › Data

DATA SET

Zimbabwe Round 10 data (2024)

18 Nov 2025

Download SAV ↓ Download CSV ↓

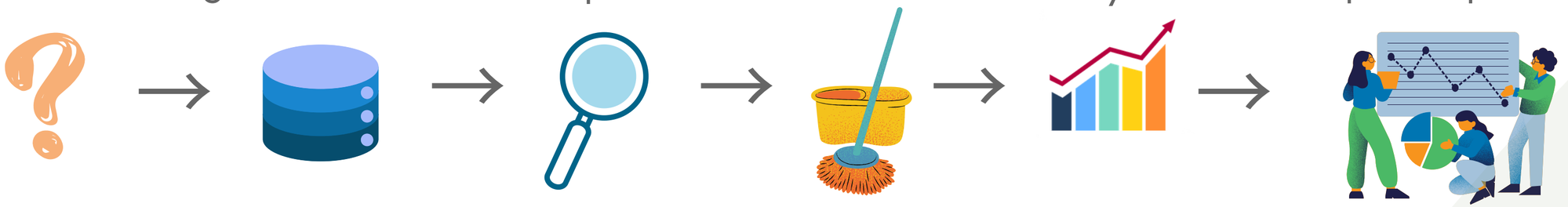
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Key steps in data analysis

- Data analysis is defined as a process of cleaning, transforming, and modeling data to discover useful information decision-making.

- The purpose of Data Analysis is to extract useful information from data and making decision based upon the data analysis.

Question → gather data → explore → clean → analyse → interpret & present



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Types of variables

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What is a variable?

- Variables are like the raw materials in quantitative analysis e.g. diabetes status, income, and smoking.
- We must be readily able to tell which variable is of what type, as misidentification of the raw materials will result in incorrect outputs. (examples?)
- Statistical methods are almost always specific to the types of variables. (e.g. linear regression when the outcome is numeric, binary logistic when the outcome is dichotomous)

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How many variables are there in the title?

Contents lists available at ScienceDirect

Journal of International Economics

journal homepage: www.elsevier.com/locate/jie



On the relationship between quality and productivity: Evidence from China's accession to the WTO☆

Haichao Fan[a], Yao Amber Li[b], Stephen R. Yeaple[c,*,1]

[a] Institute of World Economy, School of Economics, Fudan University, Shanghai, China
[b] Department of Economics and Faculty Associate of the Institute for Emerging Market Studies (IEMS), Hong Kong University of Science and Technology, Clear Water Bay, Kowloon, Hong Kong SAR, China
[c] Department of Economics, Pennsylvania State University, United States

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ABSTRACT

This paper presents an analysis of the effect of China's entry into the WTO on the quality choices of Chinese exporters in terms of their outputs and their inputs. Using highly disaggregated firm-level data, we show that the quality upgrading made possible by China's tariff reductions was concentrated in the least productive Chinese exporters. These firms, which had been laggards in terms of quality prior to the tariff reduction, were the most aggressive in increasing the quality of their exports and their inputs and in redirecting their exports toward high income markets where demand for high quality goods is strong. Our empirical results are consistent with a simple model featuring scale effect and non-Hicks' neutral productivity that disproportionately affects the efficiency with which firms use intermediate inputs. This latter feature does not appear in workhorse models of firm heterogeneity and endogenous quality choice which provide a distorted view of the impact of trade liberalization on quality upgrading.

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How many variables are there in the title?

Research article

Effects of microcredit programs on household financial portfolios: Evidence from rural Vietnam

Thi Hai Au Nguyen [a,b,*], Quoc Trung Trinh [c], Thi Loan Nguyen [d]

[a] University of Labour and Social Affairs - Ho Chi Minh City Campus, Viet Nam
[b] Ho Chi Minh City Open University, Viet Nam
[c] Ho Chi Minh City University of Law, Viet Nam
[d] Ho Chi Minh University of Banking, Viet Nam

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Rural Vietnam

ABSTRACT

Although microcredit is generally regarded as an excellent instrument for assisting borrowers in improving their well-being, its contribution to improving family financial status is often overlooked. This study addresses this issue by examining the effects of microcredit programs in Vietnam on household financial portfolios. We utilize a balanced panel dataset of 1157 identical households from two Vietnamese provinces collected in 2016, 2017, and 2022. The estimated results are calculated using the multivariate Tobit method and the instrumental variable regression method. The findings prove that microcredit increases households' lending, savings, and investments. The results are consistent with the conceptual framework and robust to different econometric models and samples. The study suggests some policy implications based on its findings.

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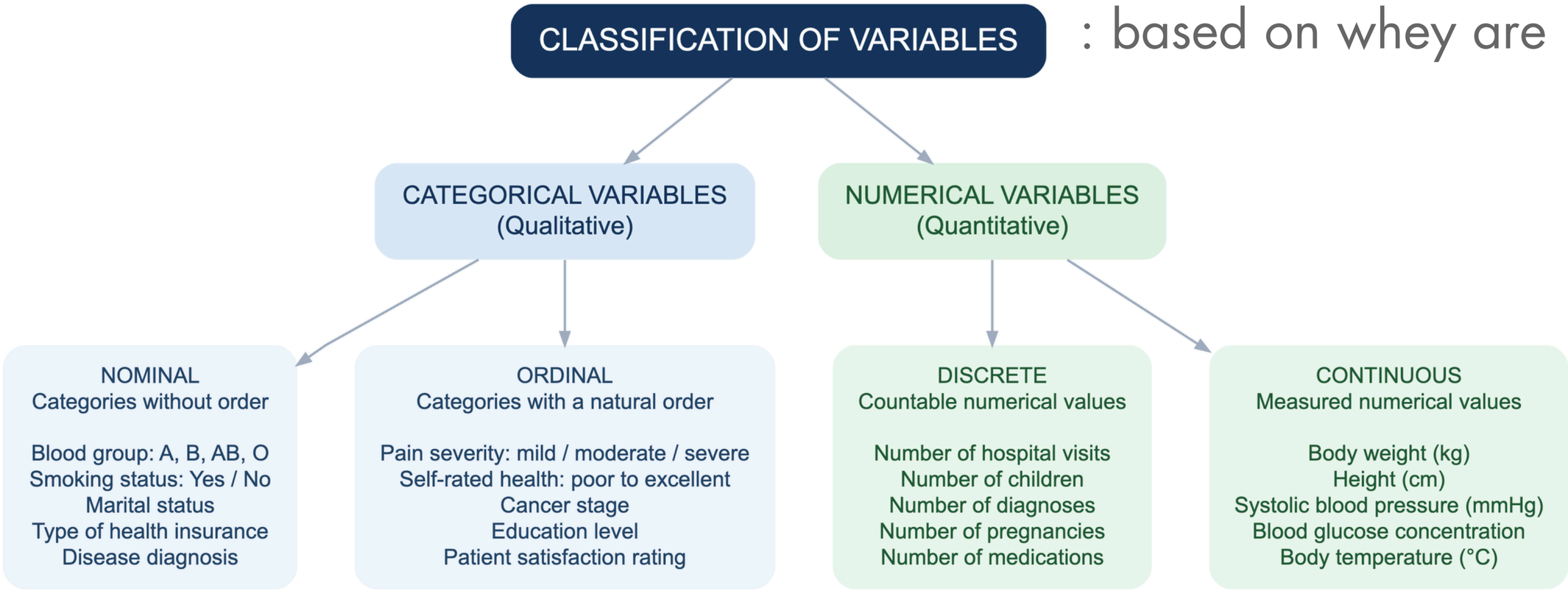
Two ways to classify variables

based on what they are made of:
e.g. numeric and non-numeric (categorical)

based on what they do:
e.g. dependent (outcome), independent (explanatory, exposure),

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CLASSIFICATION OF VARIABLES : based on whey are

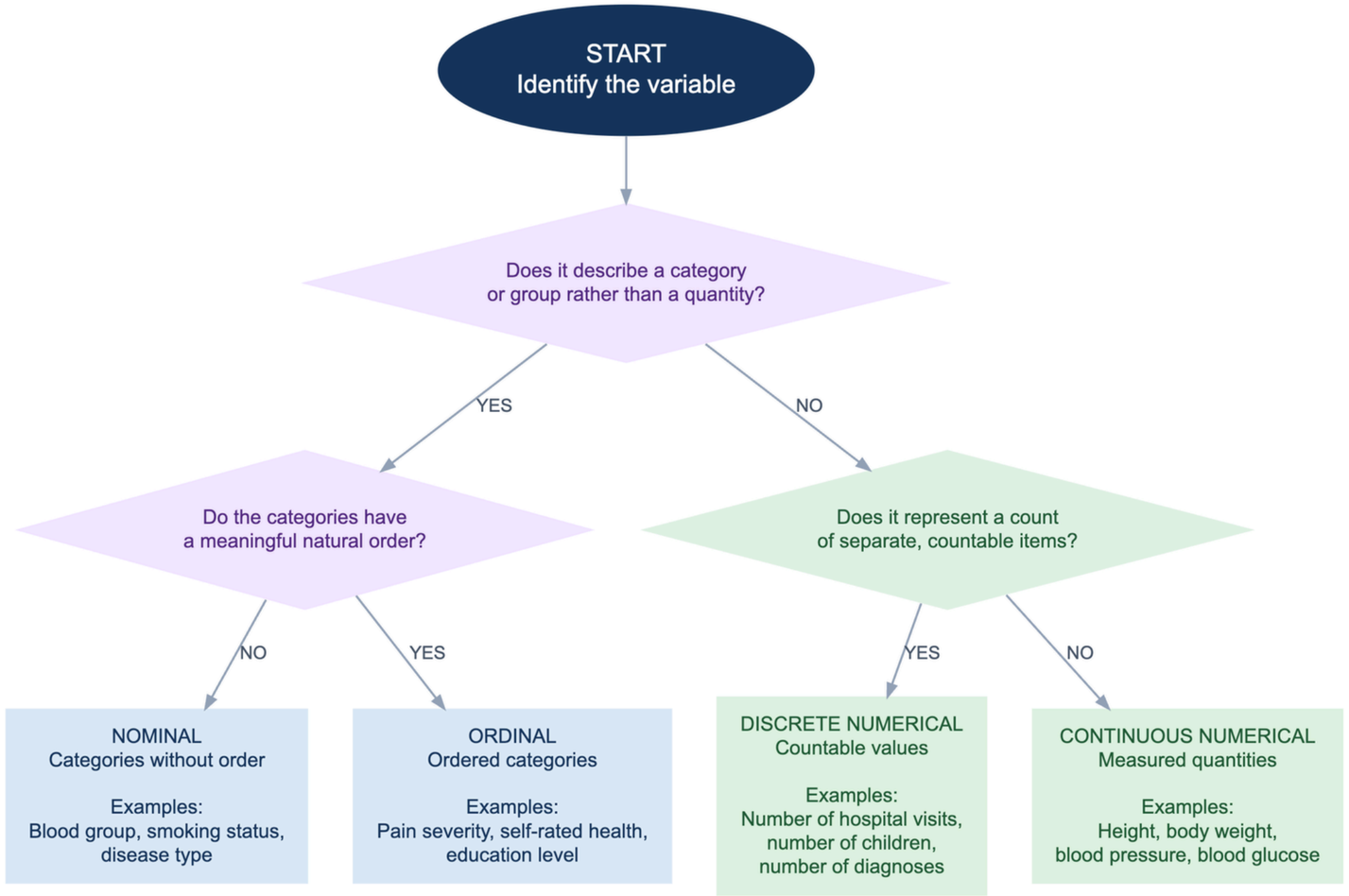


Binomial (2 categories)
Multinomial (>2 categories)

think of 5 examples of each type

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N.B.
A variable's type depends on how it is measured.
For example, age in completed years is often treated as discrete;
age measured precisely can be treated as continuous.



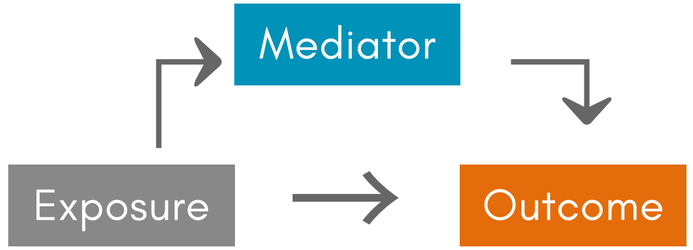
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Classification of variables: based on they do

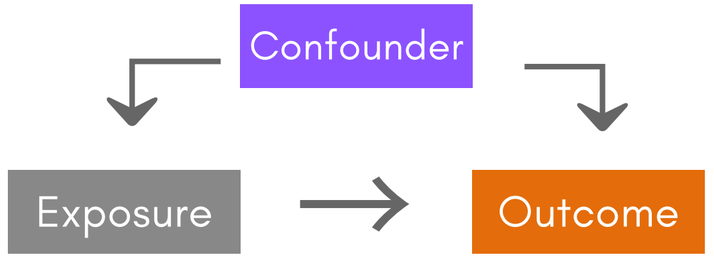
a)



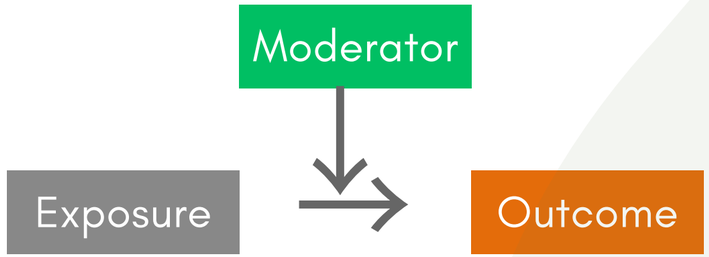
b)



c)



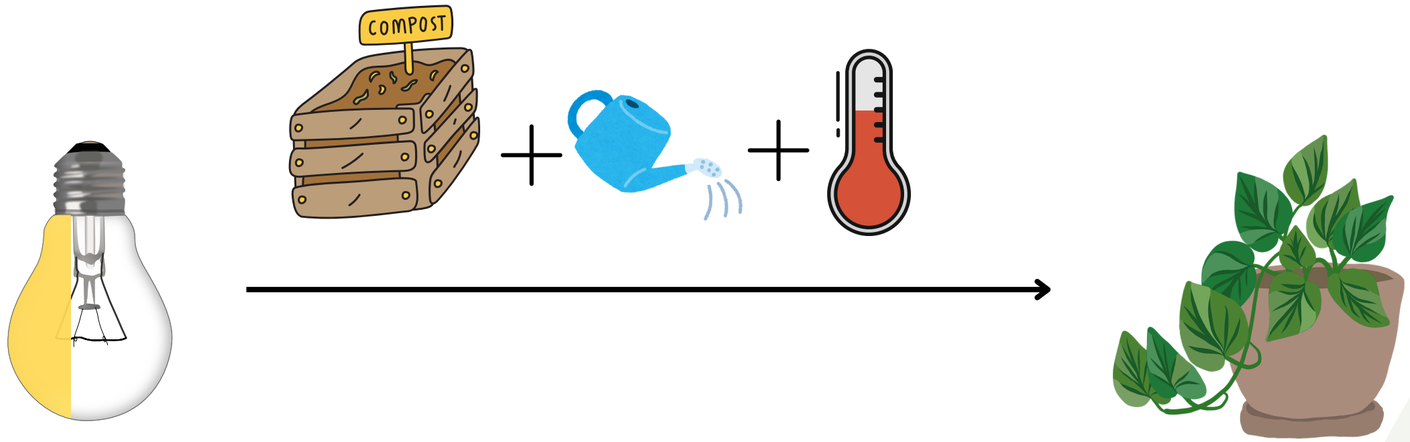
d)



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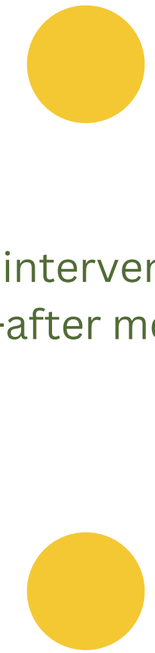
Dependent, independent & control variables

Does yellow light affect plant growth?



Types of study designs

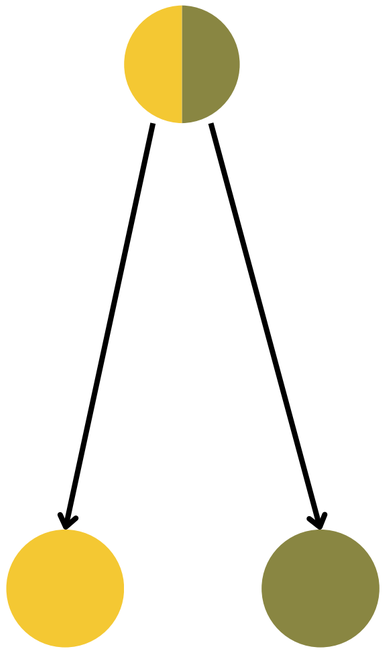
Observational studies



No intervention
no before-after measurement

Experimental studies

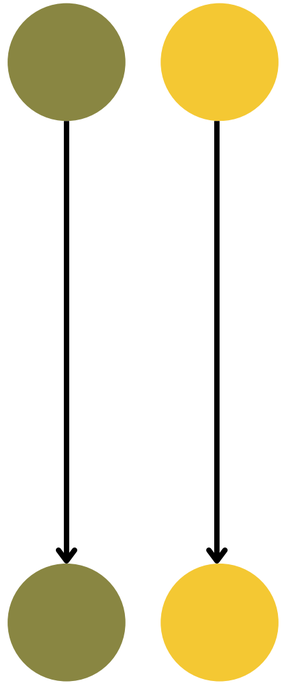
Pre-intervention



Post-intervention

Quasi-experimental

Before a natural intervention



After a natural intervention

Observational studies

- Observational studies are research processes where researchers examine subjects or phenomena without **manipulating** any variables.
- We observe and collect data on the subjects in their natural environment or context, without intervening or altering any conditions.
- Useful for generating hypotheses and studying phenomena that cannot be ethically or practically manipulated.

Experimental studies

- Experimental studies involve the intentional introduction of a treatment, procedure, or program, to observe its effects on the outcome.
- Designed to establish **cause-and-effect** relationships between variables.
- Manipulation of one or more independent variables

Does the installation of blue lights on train platforms prevent suicide? A before-and-after observational study from Japan

Tetsuya Matsubayashi [a,1], Yasuyuki Sawada [b,1], Michiko Ueda [c,*,1]

[a] University of North Texas, USA
[b] The University of Tokyo, Japan
[c] Syracuse University, 100 Eggers Hall, Syracuse, NY 13244, USA

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Suicide prevention
Blue lights

ABSTRACT

Background: Railway and metro suicides constitute a major problem in many parts of the world. Japan has experienced an increase in the number of suicides by persons diving in front of an oncoming train in the last several years. Some major railway operators in Japan have begun installing blue light-emitting-diode (LED) lamps on railway platforms and at railway crossings as a method of deterring suicides, which is less costly than installing platform screen doors. However, the effectiveness of the blue lights in this regard has not yet been proven.
Methods: This study evaluates the effect of blue lights on the number of suicides at 71 train stations by using panel data between 2000 and 2010 from a railway company in a metropolitan area of Japan. We use a regression model and compare the number of suicides before and after and with and without the intervention by the blue light. We used the number of suicides at 11 stations with the intervention as the treatment group and at the other 60 stations without the intervention as the control group.
Results: Our regression analysis shows that the introduction of blue lights resulted in a 84% decrease in the number of suicides (CI: 14–97%).
Limitation: The analysis relies on data from a single railroad company and it does not examine the underlying suicide-mitigation mechanism of blue lights.
Conclusion: As blue lights are easier and less expensive to install than platform screen doors, they can be a cost-effective method for suicide prevention.

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Types of analytical methods

Descriptive analysis

Descriptive analysis is the process of summarizing and organizing data to describe the main characteristics, patterns, or trends (not making inferences about a larger population)

- Purpose: Summarizes and describes the key features of the sample population.
- Scope: Focuses solely on the data at hand, without extending findings beyond the sample.
- Generalization: Does not generalize to a larger population.
- Uncertainty: Does not involve probability or uncertainty, as it describes only what is present.
- Example: Trend in corn production in Ontario.

Inferential test/analysis

Inferential anaysis is one whereby inference is made about the population based upon analysis of a sample selected from that population.

- Purpose: Makes inferences and generalizations about a population
- Scope: Uses sample data to make predictions about a larger population
- Generalization: Extends findings from the sample to the population
- Uncertainty: Accounts for uncertainty with probability measures
- Example: Test whether a new organic fertilizer increases corn yield.

Hypothesis tests

Hypothesis testing is a specific type of inference test used to evaluate whether a certain hypothesis about a population parameter (like a mean, proportion, or difference) is true or false based on sample data. These tests provide a way to evaluate whether observed data are likely to occur under certain assumptions.

It involves setting up a null hypothesis (H_0) and an alternative hypothesis (H_1), then using sample data to determine whether there's enough evidence to reject the null hypothesis in favor of the alternative.

Common hypothesis tests

t-test: Tests whether the means of two groups are significantly different from each other. Example: Testing if the average height of golf players differs significantly from that of football players.

z-test: Tests whether there is a significant difference between the means of two groups or to compare a sample mean to a known population mean when the sample size is large (typically n > 30) and when the population standard deviation is known.

ANOVA (Analysis of Variance): Tests if the means of three or more groups are significantly different. Example: Testing whether the average height of golf players, football players, and hockey players differs significantly across the three sports.

Chi-square test: Tests the relationship between two categorical variables.

Example: Testing if there is an association between diabetes (Yes/No) and smoking status(Yes/No).

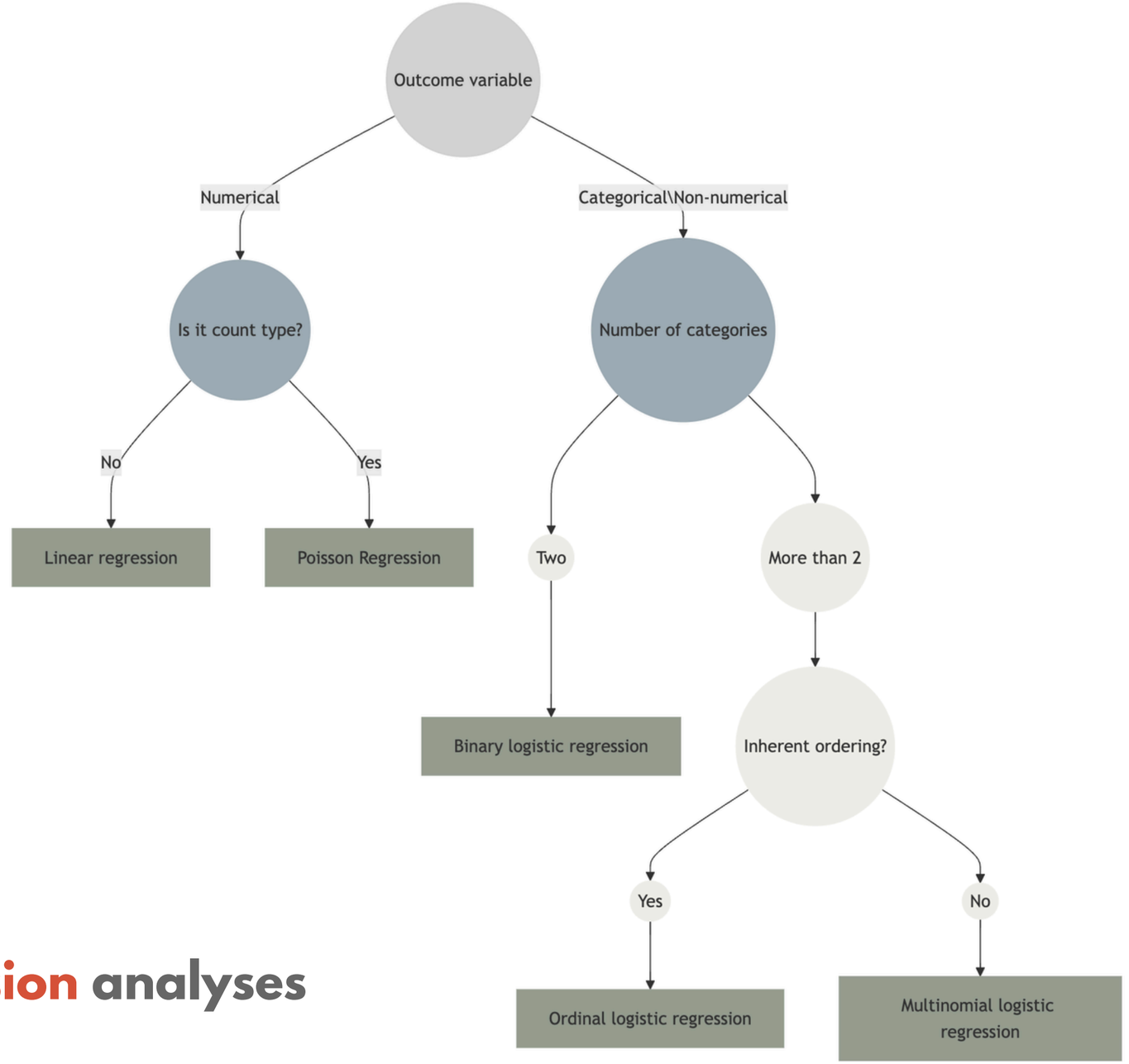
Chi-Square (χ^2) test

Chi-2 test is used to determine if there is a significant association between two categorical variables e.g. exercise habits (active vs. inactive) and smoking status (smoker vs. non-smoker). It compares the observed frequencies in each category to the expected frequencies if there were no association.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

O = Observed frequency (actual count in the data)
E = Expected frequency (theoretical count if there were no association)

Sometimes >1 possible methods
Depends of the type of DV.



Classification tree of regression analyses

Let’s practice some of the methods

install R first, then R studio Downloading of the software and packages require an Internet connection



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questions?

THANK YOU!